

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$a(3 - x) - b = -1 - 2x$$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?

- Answer**
- A. $a = 2$ and $b = 1$
- B. $a = 2$ and $b = 7$
- C. $a = -2$ and $b = 5$
- D. $a = -2$ and $b = -5$

Correct Answer: B

Rationale

Choice B is correct. Distributing the a on the left-hand side of the equation gives $3a - b - ax = -1 - 2x$. Rearranging the terms in each side of the equation yields $-ax + 3a - b = -2x - 1$. Since the equation has infinitely many solutions, it follows that the coefficients of x and the free terms on both sides must be equal. That is, $-a = -2$, or $a = 2$, and $3a - b = -1$. Substituting 2 for a in the equation $3a - b = -1$ gives $3(2) - b = -1$, so $b = 7$.

Choice A is incorrect and may be the result of a conceptual error when finding the value of b. Choices C and D are incorrect and may result from making a sign error when simplifying.